

PENDINE, United KingdomWMO: **036080**Lat: **51.750N**Lon: **4.517W**Elev: **16**StdP: **14.69**Time Zone: **0.00 (EUW)**Period: **88-99**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.8	30.0	24.0	18.2	33.9	25.8	19.8	34.7	30.2	49.4	28.0	49.0	4.5	50	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	74.8	64.1	71.5	62.3	68.7	61.4	66.0	71.6	64.2	68.8	62.8	66.9	6.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	88.8	68.1	62.4	84.3	66.1	61.0	80.1	64.4	30.7	71.1	29.3	68.6	28.3	66.8	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.0	22.6	20.0	DB	24.3	78.6	2.4	3.2	22.5	81.0	21.1	82.8	19.8	84.7	18.0	87.0
			WB	23.4	68.2	3.2	2.7	21.1	70.1	19.2	71.7	17.5	73.2	15.1	75.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	42.4	42.6	45.1	47.0	52.7	56.3	60.9	60.7	56.4	52.1	46.5	41.9	
	DBStd	8.18	5.04	5.04	3.83	4.19	5.09	3.71	3.72	4.37	3.29	5.08	5.34	5.80	
	HDD50	1162	237	208	155	107	28	2	0	0	1	38	130	256	
	HDD65	5342	702	627	616	541	381	262	137	148	258	399	556	715	
	CDD50	1320	1	1	4	16	112	192	337	330	193	104	24	6	
	CDD65	26	0	0	0	0	1	2	9	14	0	0	0	0	
	CDH74	139	0	0	0	0	2	6	48	83	0	0	0	0	
	CDH80	16	0	0	0	0	0	0	2	14	0	0	0	0	
Wind	WSAvg	9.0	10.1	10.0	9.3	9.1	8.2	8.7	7.5	7.6	9.1	10.1	8.5	9.5	
Precipitation	PrecAvg	52.3	6.1	4.4	3.2	3.3	3.1	2.9	3.0	4.0	3.6	5.9	6.5	6.2	
	PrecMax	63.9	10.0	10.2	5.9	6.6	7.4	7.0	6.0	9.6	6.5	10.8	11.1	11.9	
	PrecMin	44.0	0.4	1.2	0.9	0.8	0.4	0.7	0.8	0.3	0.6	1.7	4.1	1.1	
	PrecStd	5.7	2.5	2.3	1.6	1.5	1.6	1.8	1.3	2.2	1.4	2.3	2.1	2.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	53.0	57.2	63.0	73.8	73.8	78.7	81.5	70.1	65.3	58.0	54.8	
		MCWB	50.1	49.1	51.3	53.2	61.7	61.9	64.6	68.1	62.3	59.6	55.1	53.1	
	2%	DB	51.2	51.0	53.5	58.5	69.2	70.2	74.6	75.5	66.6	62.2	56.6	53.8	
		MCWB	49.4	48.6	49.3	51.7	59.9	60.6	63.5	66.2	60.1	58.5	54.9	52.2	
	5%	DB	50.3	50.1	51.8	55.5	65.7	66.3	71.4	70.8	64.8	60.6	55.5	52.6	
		MCWB	48.7	48.0	48.3	50.1	58.3	57.8	62.1	63.5	59.3	57.8	53.7	50.7	
	10%	DB	49.5	49.1	50.6	53.4	61.9	63.3	68.2	68.0	63.0	59.4	54.4	51.2	
		MCWB	47.9	47.1	47.7	49.0	55.2	56.9	61.2	62.2	58.6	56.6	52.6	49.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	51.0	52.3	56.0	63.4	64.0	67.6	69.1	64.1	61.3	56.7	54.0
			MCDB	51.6	51.7	54.6	62.1	71.3	70.5	75.6	78.0	67.4	63.4	57.0	54.4
2%		WB	50.2	49.5	50.8	52.9	60.4	61.3	65.3	66.9	62.1	59.7	55.4	53.0	
		MCDB	50.8	50.4	52.6	56.4	67.9	67.9	71.1	73.2	64.8	61.1	56.1	53.7	
5%		WB	49.4	48.5	49.6	51.2	58.3	59.4	63.7	65.2	60.9	58.5	54.3	51.3	
		MCDB	50.2	49.7	51.1	54.1	64.7	64.2	68.6	69.6	63.0	60.1	55.2	52.4	
10%		WB	48.5	47.7	48.6	50.0	56.3	58.1	62.3	63.4	59.7	57.3	53.1	49.8	
		MCDB	49.5	49.0	50.0	52.3	61.1	61.8	66.6	67.0	62.2	59.2	54.3	51.1	
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.9	9.9	12.1	13.7	12.1	13.0	13.0	12.5	10.8	10.3	9.0	
		MCDBR	8.7	8.9	10.6	14.0	19.4	19.0	19.4	17.1	15.2	10.3	8.6	7.3	
		MCWBR	8.7	7.5	7.9	9.1	11.9	11.5	10.9	10.2	10.2	8.1	7.5	7.8	
	5% WB	MCDBR	8.6	7.6	9.3	12.1	17.7	14.8	15.3	15.4	13.1	9.6	7.3	7.1	
		MCWBR	8.9	7.3	7.9	9.0	11.1	9.8	9.7	9.9	10.2	8.6	7.0	7.8	
Clear-Sky Solar Irradiance	taub	0.334	0.346	0.375	0.389	0.397	0.385	0.389	0.372	0.366	0.356	0.332	0.324		
	taud	2.324	2.368	2.309	2.296	2.319	2.382	2.378	2.440	2.447	2.430	2.379	2.302		
	Ebn at Noon	208	244	258	269	272	276	272	272	260	237	209	190		
	Edh at Noon	21	26	33	37	38	36	36	32	29	24	19	18		
All-Sky Solar Radiation	RadAvg	234	479	872	1392	1711	1860	1695	1416	1046	580	291	176		
	RadStd	23	43	75	98	108	157	141	94	76	47	24	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page